Prabhav Pradeep

ch24btech11029@iith.ac.in | prabhavpradeep046@gmail.com | +91 8089455076 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.09
Kendriya Vidyalaya (DRDO) Kanchanbagh	Class XII (CBSE)	2024	96.60%
PM SHRI Kendriya Vidyalaya No. 1 Calicut	Class X (CBSE)	2022	96.60%

POSITIONS OF RESPONSIBILITY

Consulting Coordinator | Finance and Consulting Club, IIT Hyderabad

- Drove **5,000+** footfall for Horizon, IIT Hyderabad's flagship management fest, by leading a 10-member associate team to architect high-engagement competitions and curated event experiences.
- Mobilised **4,000+** participants from across India by orchestrating Prodx, a national product case competition.
- Co-authored the first edition of the **FCC PrepBook**, bridging the preparation gap for **5,000+** students campus-wide with **40+ case interview transcripts** and a structured resource for management interview preparation.
- Ideated and conducted **8+ sessions** on product and consulting frameworks and cases, drawing **200+ students**

TEDxIIT Hyderabad | Content and Curation

- **Curated talk concepts and refined messaging** for **10+** guest speakers, including international athletes and award-winning actors, helping them structure their raw ideas into engaging talks for **1,500+** attendees.
- **Served as the primary Point of Contact** for senior **IPS officer Shri P.M. Nair**, collaborating over **10+** sessions to extract key public service themes and structure his core insights into a well-paced, cohesive address.
- **Authored a 30-page comprehensive study guide** for xMUN'25 on *"Combating Terror Financing via Cryptocurrency,"* conducting extensive policy research to prepare **50+ UNGA delegates** for committee proceedings.
- **Managed UNGA live Press Corps operations**, leading a 4-member media team to cover real-time policy debates and author **2** detailed committee op-eds.
- **Collaborated with 10-member core team** to brainstorm and conceptualize the flagship annual theme and strategize the overarching marketing rollout, contributing to the foundational planning and direction of the main event.

SKILLS

Product & Project Management: Jira, Asana, Agile/Scrum, CIRCLES Framework, Go-To-Market (GTM) Strategy, MVP Scoping

Tools/Frameworks: NumPy, Pandas, Scikit-learn, OpenCV, TensorFlow, LightGBM, SHAP, SciPy, FAISS, Rank-BM25, RAGAS, LangChain, LangGraph

Domains: Machine Learning, Deep Learning, Computer Vision, Data Science

RELEVANT COURSES

Humanities & Management: Entrepreneurship, Organization Structure, Disability Studies

Mathematics and Statistics: Applied Mathematics, Numerical Methods, Linear Algebra, Differential Equations, Calculus

Computer Science: Introduction to Programming, Artificial Intelligence

PROJECTS

RTB Ad Exchange Simulator -- Marketplace Auction Dynamics || (Python,FastAPI,asyncio) | [Link]

- **Built** a distributed RTB ad exchange fanning out concurrent bid requests to **3 DSPs under a 100ms SLA**
- **Diagnosed** a connection-pooling bug causing false bidder timeouts under load; **fixed it, halving p50 latency**

- **Implemented OpenRTB-compliant** first- and second-price auction clearing with schema and floor enforcement
- **Containerized** with **Docker** and deployed on **AWS EC2, with CI/CD** automation via GitHub Actions

Scalable Financial RAG Pipeline & API || (FastAPI, FAISS, Ollama, RAGAS) | [Link]

- **Built** a deployable, end-to-end **RAG architecture** parsing complex **SEC filings**, integrating hybrid **FAISS/BM25** retrieval and **Cross-Encoder reranking** to maximize context precision.
- **Wrapped** the retrieval and local Llama 3.2 generation layers into an asynchronous **FastAPI REST server**, designing scalable endpoints for dynamic financial query processing.
- **Quantified** pipeline reliability via automated LLM-as-a-Judge evaluation (**RAGAS**), benchmarking system guardrails against **Faithfulness and Answer Relevancy** metrics for production readiness.

Chronos -- Agentic Scheduling Assistant || (Python, LangGraph, Google Calendar API) | [Link]

- **Built** a **LangGraph-based agent** that classifies unstructured input (**parsed emails/text**), routes between event-creation and multi-step goal-decomposition branches, and auto-writes to a live calendar
- **Designed** a goal-decomposition flow that interviews the user for subtasks on long-horizon commitments (e.g. exams) and schedules each into open calendar slots via a first-fit slot-finding algorithm
- **Added** human-in-the-loop confirmation via LangGraph interrupts before any calendar write, guarding against agent scheduling errors

Zudio Changing Room Redesign -- Product Case Study || (MVP Design)

- Diagnosed changing-room **UX friction** via first-person research; used **CIRCLES** to assess accessibility issues.
- Designed a **low-capex** pilot using repurposed rooms and a priority queue, with metrics to validate demand pre-redesign.

Telco Customer Churn Prediction || (Python, Scikit-learn, LightGBM, SHAP, Streamlit)

- Built a cost-sensitive churn pipeline comparing Logistic Regression, Random Forest, LightGBM, and a **soft-voting ensemble** with custom cost-optimal thresholding (5:1 FN:FP ratio).
- Deployed LightGBM **(PR-AUC 0.64, 91% recall**) with live **SHAP waterfall** explanations via Streamlit.

ACADEMICS

Secured a **9.59 GPA** across Semesters II & III (joint-highest in Chemical Engineering), outperforming the top academic standings across seven distinct engineering departments

- **Scored the highest in class** in 4 core subjects (**Heat Transfer, Material Science, Applied Mathematics**), including a **perfect 100/100** in **Transport Phenomena.**
- **Second Major in Electrical Engineering:** Focused on **semiconductor technology** and the cross-functional application of **electrical and chemical engineering systems.**
- **Elected Core Coursework: Computer Vision** (applied to ChemE), **Computational Biology** and **Polymer Rheology**.
- **Japanese Learning**: Currently taking **Spoken Japanese Basics(LA1210)** and planning to obtain **JLPT-N5 and further.**

CERTIFICATIONS

Google Project Management Specialization -- Coursera (Covering Agile/Scrum, Project Planning, Risk Management & Asana)

- Participated in various hackathons like, EY Techathon 6.0, ProduScope 2025, Amex Strategy Hackathon 2026
- Completed free versions of various courses like Stanford CS229, Agentic AI, through Deeplearning.ai

HOBBIES AND INTERESTS

- **Interests:** Japanese Media & Culture (**Anime, Manga**) and board games like **Chess and Checkers**

SCHOLASTIC ACHIEVEMENTS

NDA & NA(II) 2023: Secured **top 1.3%** among **6,00,000+** candidates nationally (7,416 selected)

KVS Merit Certificate (Class X, 2022): Ranked in the **top 1.5%** of **100,000+** KV students nationwide.

KVS Kanchanbagh Hyderabad: Secured **1st position** in school, CBSE Class XII, 2024